

6th Semester

Geophysical Inverse Theory (EX50017)

3-1-0 Total credit-4

Introduction: Basic definition of inversion, observational data, model parameters, Definition of the forward and inverse problems, linear dependence and independence, existence, constructibility, approximation, stability and nonuniqueness of an inverse problem, Ill-posed and well-posed problem, Term project Introduction. (*Lectures 1-6*)

Matlab Sessions with examples: Formulation of inverse problems: implicit and explicit linear forms, Introduction to Seismic tomography. (*Lectures 7-9*)

Introduction of Probability and Statistics: Elements of probability theory and principles of Bayesian inversion, Variance, Covariance, Correlation Matrix, Conditional, Marginal and Joint Probability Density Function, example of Bayesian inversion. (*Lectures 10-12*)

Linear Discrete Inverse Problems: Overdetermined, Even-determined and Underdetermined problem; Solutions of linear inverse problem, vector norms, prediction error, solution of the Overdetermined problem, example of Overdetermined problem and Matlab session, priori information, constraints, Lagrange Multiplier, Minimum Norm Algorithm for an Underdetermined Problem, example of Underdetermined problem and Matlab session, Mixed-Determined problems, Weighted Least squares, Weighted Minimum length. (*Lectures 13-18*)

Generalized inverse: The data resolution, the model resolution, unit covariance matrix, Measures of goodness of resolution and covariance, few examples, Sidelobes and the Backus-Gilbert Spread function. (*Lectures 19-21*)

Singular value decomposition: Eigenvalues and Eigenvector, decomposition of an inverse problem, Properties of SVD, Example of SVD (*Lectures 22-24*)

Term Project Session-I: Discussion of the assignments. (*Lectures 25-27*)

Non-linear problems: Basic concept with examples, Quasi-linear problem, Global Optimization, Monte Carlo Inversion, Resolution and example, Simulated Annealing with examples, Genetic Algorithm with examples, Brief Introduction to Neural Networks and Deep Learning. (*Lectures 28-38*)

Term Project Session-II: Group project presentations. (*Lectures 39-42*).

Texts and Refs (Should have classical as well as recent):

1. Geophysical Data Analysis, by William Menke, Academic Press (Elsevier) 1989.
2. Understanding inverse theory, Parker, R. L. (1977), Ann. Rev. Earth Planet. Sci., 5, 35-64.
3. Parameter Estimation and Inverse Problems, by Richard Aster, Brian Borchers, and Clifford Thurber, Academic Press (Elsevier) 2004

4. Introductory Geophysical Inverse Theory by John A. Scales, Martin L. Smith and Sven Treitel.
5. Geophysical Inverse Theory, by Robert Parker, Princeton University Press, 1994.
6. Potential Theory in Applied Geophysics, by K. K. Roy
7. Global Optimization Methods in Geophysical Inversion by M. Sen and P. L. Stoffa (Elsevier).
8. *Monte Carlo methods in geophysical inverse problems*, Sambridge, M. and K. Mosegaard (2002), Rev. of Geophys., 40, 3.1-3.29.

Teachers:

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